

WEEK 1: Introduction to Mechanical Engineering, Fundamental & Derived units

Introduction to the discipline, career pathways, and the fundamental & derived units used in engineering calculations

What is Mechanical Engineering?

Mechanical Engineering is the study of machines and how they work. It involves designing, making, operating, and repairing machines used in industries and daily life..

KEY IDEA

It is often called the 'mother engineering' branch — the core knowledge behind almost every machine humans build.



Design

Concept to CAD models



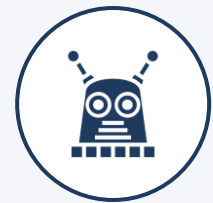
Thermal / Energy

Heat, power & fluid systems



Manufacturing

Production & materials



Automation

Robotics & mechatronics

Scope of Mechanical Engineering

A discipline with applications across nearly every industry



Automotive

Engine design, EVs, vehicle dynamics



Aerospace

Propulsion, structures, avionics



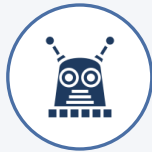
Manufacturing

Production, CNC, quality control



Energy & Power

Turbines, HVAC, renewables



Robotics & Mechatronics

Automation & smart systems



R&D and Materials

Product design, simulation, FEA

Career Opportunities for Mechanical Engineers



Design & Production Engineer

Automotive, manufacturing & core industries



R&D and Product Development

Innovation, simulation, prototyping



Government & PSU Jobs

GATE / ESE, railways, defence



Higher Studies

B.E./B.Tech, M.Tech, MS abroad



Global Opportunities

Oil & gas, aerospace, marine sectors



Entrepreneurship

Startups in automation & robotics

Fundamental Units – Definition

Fundamental (Base) Units are units of measurement that are independent of one another and cannot be derived from any other unit. They form the foundation for all derived units.



Length

CGS	centimetre (cm)
FPS	foot (ft)
MKS	metre (m)
SI	metre (m)



Mass

CGS	gram (g)
FPS	pound (lb)
MKS	kilogram (kg)
SI	kilogram (kg)



Time

CGS	second (s)
FPS	second (s)
MKS	second (s)
SI	second (s)



Temperature

CGS	°Celsius
FPS	°Fahrenheit
MKS	°Celsius / Kelvin
SI	Kelvin (K)

1. CGS System

Full Form: Centimetre–Gram–Second

Definition:

The CGS system uses centimetre (cm) for length, gram (g) for mass, and second (s) for time.

Fundamental Quantity

Unit

Length

Centimetre (cm)

Mass

Gram (g)

Time

Second (s)

Examples

- Length of a pencil = 15 cm
- Mass of a chocolate = 50 g
- Time to blink = 1 s

Used In

- Older physics books
- Small laboratory measurements

2. FPS System

Full Form: Foot–Pound–Second

Definition:

The FPS system uses foot (ft) for length, pound (lb) for mass (or weight in common usage), and second (s) for time.

Fundamental Quantity	Unit
Length	Foot (ft)
Mass	Pound (lb)
Time	Second (s)

Examples

- Height of a person = 6 ft
- Weight of a suitcase = 20 lb
- Time = 30 s

Used In

- Some countries like the United States for everyday measurements.

3. MKS System

Full Form: Metre–Kilogram–Second

Definition:

The MKS system uses metre (m) for length, kilogram (kg) for mass, and second (s) for time.

Fundamental Quantity	Unit
Length	Metre (m)
Mass	Kilogram (kg)
Time	Second (s)

Examples

- Length of a classroom = 8 m
- Mass of a bag of rice = 10 kg
- Time = 60 s

Used In

- Engineering
- Science

4. SI System

Full Form: International System of Units (French: *Système International d'Unités*)

Definition:

The SI system is the internationally accepted standard system of measurement used all over the world.

Seven SI Fundamental Units

Physical Quantity	SI Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Temperature	kelvin	K
Amount of Substance	mole	mol
Luminous Intensity	candela	cd

Examples

- Height of a student = 1.6 m
- Mass of a person = 60 kg
- Room temperature = 300 K

Basic Units and Conversions (SI System)

1. Length

- 1 kilometre (km) = 1000 metres (m)
- 1 metre (m) = 100 centimetres (cm)
- 1 centimetre (cm) = 10 millimetres (mm)

2. Mass

- 1 tonne = 1000 kilograms (kg)
- 1 kilogram (kg) = 1000 grams (g)
- 1 gram (g) = 1000 milligrams (mg)

3. Time

- 1 hour = 60 minutes
- 1 minute = 60 seconds
- 1 day = 24 hours

4. Volume

- 1 litre (L) = 1000 millilitres (mL)
- 1 cubic metre (m³) = 1000 litres

5. Temperature

- $^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$
- $^{\circ}\text{F} = (^{\circ}\text{C} \times 9/5) + 32$

6. Speed

- Speed = Distance $\div$ Time
- 1 m/s = 3.6 km/h
- 1 km/h = 0.2778 m/s

7. Area

- 1 m² = 10,000 cm²
- 1 hectare = 10,000 m²

8. Force

- Unit: Newton (N)
- 1 N = 1 kg·m/s²

Derived Units – Definition

Derived Units

A **derived unit** is a unit that is **made by combining two or more fundamental (base) units** using multiplication or division.

Example – how Force is derived:

$$\text{Force} = \text{Mass} \times \text{Acceleration}$$

$$\text{Acceleration} = \text{Length} \div \text{Time}^2$$

$$\text{So Force unit} = \text{kg} \cdot \text{m} / \text{s}^2 = \text{newton (N)}$$

$$\text{Speed} = \text{Distance} \div \text{Time}$$

$$\text{Unit of Speed} = \text{metre/second (m/s)}$$

Here, **m/s** is a **derived unit** because it is formed from **metre** and **second**.



Area



Volume



Density



Velocity



Acceleration



Force



Work



Energy



Power

Derived Units – Formulae & SI Units (Part 1)

Quantity	Formula	SI Unit	CGS / FPS Unit
Area 	Length $\times$ Breadth	m ²	cm ² (CGS) / ft ² (FPS)
Volume 	Length $\times$ Breadth $\times$ Height	m ³	cm ³ (CGS) / ft ³ (FPS)
Density 	Mass / Volume	kg/m ³	g/cm ³ (CGS) / lb/ft ³ (FPS)
Velocity 	Displacement / Time	m/s	cm/s (CGS) / ft/s (FPS)
Acceleration 	Velocity / Time	m/s ²	cm/s ² (CGS) / ft/s ² (FPS)

Derived Units – Formulae & SI Units (Part 2)

Quantity	Formula	SI Unit	CGS / FPS Unit
Force 	Mass × Acceleration	newton (N) = $\text{kg}\cdot\text{m}/\text{s}^2$	dyne (CGS) / poundal (FPS)
Work 	Force × Displacement	joule (J) = $\text{N}\cdot\text{m}$	erg (CGS) / $\text{ft}\cdot\text{lb}$ (FPS)
Energy 	Capacity to do Work	joule (J)	erg (CGS) / $\text{ft}\cdot\text{lb}$ (FPS)
Power 	Work / Time	watt (W) = J/s	erg/s (CGS) / hp (FPS)

Handy conversions: $1 \text{ N} = 10^5 \text{ dyne}$ | $1 \text{ J} = 10^7 \text{ erg}$ | $1 \text{ hp} = 746 \text{ W}$ | $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

Worked Example – Force & Work



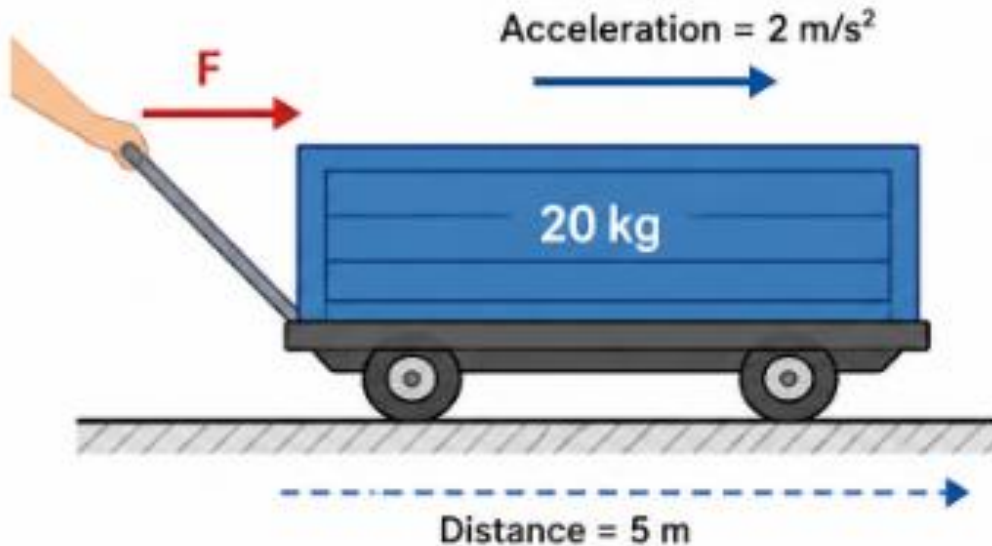
A force pushes a 20 kg trolley, giving it an acceleration of 2 m/s^2 . Find the Force applied and the Work done if the trolley moves 5 m.

Given:

Mass of trolley (m) = 20 kg

Acceleration (a) = 2 m/s^2

Distance moved (s) = 5 m



1. Force Applied

Using Newton's Second Law,

$$F = m \times a$$

$$F = 20 \text{ kg} \times 2 \text{ m/s}^2$$

$$F = 40 \text{ N}$$

2. Work Done

Work done, $W = F \times s$

$$W = 40 \text{ N} \times 5 \text{ m}$$

$$W = 200 \text{ J}$$

Answers:

- Force applied, $F = 40 \text{ N}$
- Work done, $W = 200 \text{ J}$

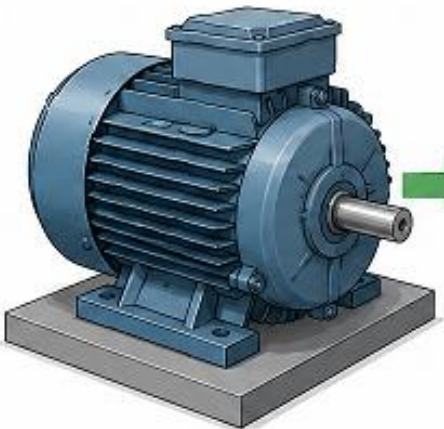
Worked Example – Power



A motor does 3000 J of work in 15 seconds. Calculate the power delivered in watts and convert it to horsepower (hp).

Given:

- Work done (W) = 3000 J
- Time (t) = 15 s



Work done
W = 3000J

Time taken
t = 15 s

1. POWER IN WATTS

$$\text{Power, } P = \frac{\text{Work}}{\text{Time}}$$

$$P = \frac{3000 \text{ J}}{15 \text{ s}} = 200 \text{ W}$$

1 watt (W) = 1 joule
of work done in
1 second.

2. CONVERT WATTS TO HORSEPOWER (hp)

$$1 \text{ hp} = 746 \text{ W}$$

$$\text{hp} = \frac{P \text{ (in W)}}{746}$$

$$\text{hp} = \frac{200}{746} = 0.268 \text{ hp}$$

Useful Information

$$1 \text{ hp} = 746 \text{ W}$$

SAMPLE PROBLEMS ON AREA AND VOLUME (MACHINES)

1. Area of a Circular Machine Pulley



Problem:

A circular pulley of a lathe machine has a radius of 15 cm. Calculate the area of the pulley face.

Given:

Radius (r) = 15 cm

Formula:

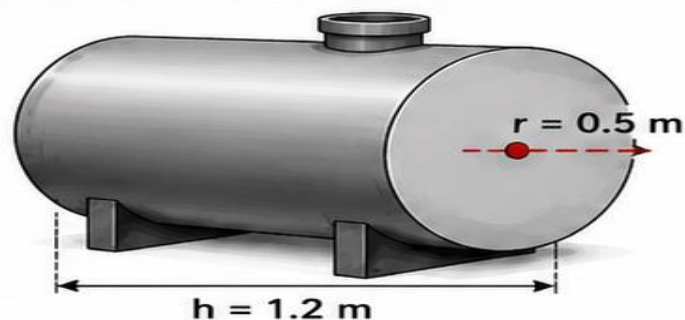
$$\text{Area of Circle} = \pi r^2$$

Solution:

$$\begin{aligned}\text{Area} &= 3.14 \times 15 \times 15 \\ &= 3.14 \times 225 \\ &= \mathbf{706.5 \text{ cm}^2}\end{aligned}$$

Answer: The area of the pulley is $\mathbf{706.5 \text{ cm}^2}$.

2. Volume of a Cylindrical Oil Tank



Problem:

A hydraulic oil tank used in a machine is cylindrical. It has a radius of 0.5 m and a height of 1.2 m. Find the volume of the tank.

Given:

- Radius (r) = 0.5 m
- Height (h) = 1.2 m

Formula:

$$\text{Volume of Cylinder} = \pi r^2 h$$

Solution:

$$\begin{aligned}\text{Volume} &= 3.14 \times (0.5)^2 \times 1.2 \\ &= 3.14 \times 0.25 \times 1.2 \\ &= \mathbf{0.942 \text{ m}^3}\end{aligned}$$

Answer: The volume of the tank is $\mathbf{0.942 \text{ m}^3}$.

3. Surface Area of a Cubical Machine Crate



Problem:

A machine spare part is packed in a cubical wooden box with each side measuring 40 cm. Calculate the total surface area of the box.

Given:

Side (s) = 40 cm

Formula:

$$\text{Surface Area of Cube} = 6s^2$$

Solution:

$$\begin{aligned}\text{Surface Area} &= 6 \times (40)^2 \\ &= 6 \times 1600 \\ &= \mathbf{9600 \text{ cm}^2}\end{aligned}$$

Answer: The total surface area of the box is $\mathbf{9600 \text{ cm}^2}$.

Note: Use $\pi = 3.14$ in all calculations.

MCQ Questions – Q1 to Q4

Test Your Understanding

1. What is Mechanical Engineering mainly concerned with?

A) Chemical reactions B) Design, manufacture & operation of machines C) Study of living organisms D) Electrical circuits

2. Which of the following is NOT a fundamental quantity?

A) Length B) Mass C) Force D) Time

3. In the SI system, the unit of length is:

A) Centimetre B) Foot C) Metre D) Inch

4. In the CGS system, the unit of mass is:

A) Kilogram B) Pound C) Gram D) Tonne

MCQ Questions – Q5 to Q8

Test Your Understanding

5. Which unit system is internationally accepted?

A) CGS B) FPS C) MKS D) SI

6. Temperature in the SI system is measured in:

A) Celsius B) Fahrenheit C) Kelvin D) Rankine

7. The SI unit of Force is:

A) Pascal B) Joule C) Newton D) Watt

8. Power is defined as:

A) Force $\times$ Distance B) Work / Time C) Mass $\times$ Velocity D) Energy $\times$ Time

MCQ Questions – Q9 to Q12

Test Your Understanding

9. The derived unit of Work / Energy in SI is:

A) Newton B) Watt C) Joule D) Pascal

10. Vernier Caliper is used to measure:

A) Temperature only B) Angles only C) Length, outer & inner diameter, depth D) Mass

11. Which instrument measures the height of an object from a datum?

A) Screw Gauge B) Height Gauge C) Vernier Caliper D) Sine Bar

12. Screw Gauge is used to measure:

A) Inside diameter B) Angles C) Outside diameter of small objects D) Height

MCQ Questions – Q13 to Q16

Test Your Understanding

13. A Bevel Protractor is used to measure:

A) Length B) Mass C) Angles D) Volume

14. The accuracy of a Vernier Caliper is typically:

A) 0.1 m B) 0.02 mm C) 1 cm D) 0.5 mm

15. 1 kilometre equals how many metres?

A) 10 m B) 100 m C) 1000 m D) 10,000 m

16. The formula to convert °F to °C is:

A) $(^{\circ}\text{F} + 32) \times 5/9$ B) $(^{\circ}\text{F} - 32) \times 5/9$ C) $(^{\circ}\text{F} \times 9/5) + 32$ D) $^{\circ}\text{F} / 1.8$

MCQ Questions – Q17 to Q20

Test Your Understanding

17. SI unit of Acceleration is:

A) m/s B) m^2/s C) m/s^2 D) $\text{kg}\cdot\text{m}/\text{s}$

18. Density is a derived unit expressed as:

A) kg/m^2 B) $\text{kg}\cdot\text{m}$ C) kg/m^3 D) m/kg

19. Sine Bar is used in workshops for:

A) Measuring mass B) Measuring precise angles C) Measuring temperature D) Measuring speed

20. In FPS system, unit of length is:

A) Metre B) Centimetre C) Foot D) Inch

MCQ Answer Key

Q1 – Q20

Q1: B

Q2: C

Q3: C

Q4: C

Q5: D

Q6: C

Q7: C

Q8: B

Q9: C

Q10: C

Q11: B

Q12: C

Q13: C

Q14: B

Q15: C

Q16: B

Q17: C

Q18: C

Q19: B

Q20: C



Review answers with your teacher after the quiz!